

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location CYS41 Datacenter, WY -- station KIBM, America/Denver
Committed pair OSM 537172525 -> 1177975783
OSM way 537172525 / OSM way 1177975783
Facade gap 184.5 m between the two halls
Weather record 43,440 real hourly records from KIBM
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.2551 C at 155 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2022-11-07 (knife_edge)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 1 mode change. The reactive incumbent took 24 hours -- 12 more -- but declared 0 unsafe hours against the agent's 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 12 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 24 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	2.461	18.0	2.885	5.495
01	mechanical	yes	switch budget	2.955	18.0	2.959	4.534
02	mechanical	yes	switch budget	3.474	18.0	1.717	4.133
03	mechanical	yes	switch budget	3.586	18.0	0.217	3.739
04	mechanical	yes	switch budget	3.749	18.0	0.217	3.519
05	mechanical	yes	switch budget	4.250	18.0	-0.393	3.582
06	mechanical	yes	switch budget	5.154	18.0	-0.688	3.400

07	mechanical	yes	switch budget	8.460	18.0	0.482	4.178
08	mechanical	yes	switch budget	11.794	18.0	1.459	5.492
09	mechanical	yes	switch budget	14.407	18.0	2.679	6.807
10	mechanical	yes	switch budget	17.146	18.0	3.973	7.562
11	mechanical	no	dry-bulb	18.225	18.0	4.619	7.714
12	FREE	yes	-	17.852	18.0	6.679	7.580
13	FREE	yes	-	16.084	18.0	8.069	6.217
14	FREE	yes	-	13.024	18.0	8.679	4.909
15	FREE	yes	-	10.668	18.0	9.289	4.330
16	FREE	yes	-	9.371	18.0	7.717	4.675
17	FREE	yes	-	7.690	18.0	7.289	4.879
18	FREE	yes	-	7.156	18.0	6.872	4.999
19	FREE	yes	-	6.121	18.0	6.863	4.867
20	FREE	yes	-	5.460	18.0	5.982	5.132
21	FREE	yes	-	6.313	18.0	5.107	5.680
22	FREE	yes	-	5.952	18.0	3.948	6.153
23	FREE	yes	-	6.303	18.0	2.852	6.142

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.461 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.955 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.474 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.586 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.749 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here

would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.249 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.154 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.460 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.794 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.407 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.146 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.225 C against a 18.0 C limit, so it fails by 0.225 C. It would take a limit 0.225 C higher, or a bound 0.225 C tighter, to change this hour.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.852 C, which is 0.148 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 7.580 C, and the margin is measured, not chosen: 7.498 C of group-conditional forecast error for this hour of day, plus 0.0818 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.085 C, which is 1.915 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 6.217 C, and the margin is measured, not chosen: 6.133 C of group-conditional forecast error for

this hour of day, plus 0.0843 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.024 C, which is 4.976 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.909 C, and the margin is measured, not chosen: 4.836 C of group-conditional forecast error for this hour of day, plus 0.0737 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.667 C, which is 7.333 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.330 C, and the margin is measured, not chosen: 4.272 C of group-conditional forecast error for this hour of day, plus 0.0575 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.371 C, which is 8.629 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.675 C, and the margin is measured, not chosen: 4.583 C of group-conditional forecast error for this hour of day, plus 0.0914 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.690 C, which is 10.310 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.879 C, and the margin is measured, not chosen: 4.793 C of group-conditional forecast error for this hour of day, plus 0.0856 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.156 C, which is 10.844 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.999 C, and the margin is measured, not chosen: 4.904 C of group-conditional forecast error for this hour of day, plus 0.0953 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.121 C, which is 11.879 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.867 C, and the margin is measured, not chosen: 4.775 C of group-conditional forecast error for this hour of day, plus 0.0921 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.460 C, which is 12.540 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.132 C, and the margin is measured, not chosen: 5.062 C of group-conditional forecast error for this hour of day, plus 0.0703 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.313 C, which is 11.687 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.680 C, and the margin is measured, not chosen: 5.587 C of group-conditional forecast error for this hour of day, plus 0.0932 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.952 C, which is 12.048 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 6.153 C, and the margin is measured, not chosen: 6.055 C of group-conditional forecast error for this hour of day, plus 0.0979 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.303 C, which is 11.697 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 6.142 C, and the margin is measured, not chosen: 6.033 C of group-conditional forecast error for this hour of day, plus 0.1087 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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